

Table 15.1A Relative magnitude of IT tolerances for grades 5 to 16 in terms of tolerance unit i for sizes upto 500 mm

Grade	IT 5	IT 6	IT 7	IT 8	IT 9	IT 10	IT 11	IT 12	IT 13	IT 14	IT 15	IT 16
Tolerance values	$7i$	$10i$	$16i$	$25i$	$40i$	$64i$	$100i$	$160i$	$250i$	$400i$	$640i$	$1000i$

Table 15.1 Fundamental tolerances of grades 01, 0 and 1 to 16 (values of tolerances in microns) (1 micron = 0.001 mm)

Diameter steps in mm	Tolerance Grades																	
	01	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14*	15*	16*
To and inc 3	0.3	0.5	0.8	1.2	2	3	4	6	10	14	25	40	60	100	140	250	400	600
Over 3																		
To and inc 6	0.4	0.6	1	1.5	2.5	4	5	8	12	18	30	48	75	120	180	300	480	750
Over 6																		
To and inc 10	0.4	0.6	1	1.5	2.5	4	6	9	15	22	36	58	90	150	220	360	580	900
Over 10																		
To and inc 18	0.5	0.8	1.2	2	3	5	8	11	18	27	43	70	110	180	270	430	700	1100
Over 18																		
To and inc 30	0.6	1	1.5	2.5	4	6	9	13	21	33	52	84	130	210	330	520	840	1300
Over 30																		
To and inc 50	0.6	1	1.5	2.5	4	7	11	16	25	39	62	100	160	250	390	620	1000	1600
Over 50																		
To and inc 80	0.8	1.2	2	3	5	8	13	19	30	46	74	120	190	300	460	740	1200	1900
Over 80																		
To and inc 120	1	1.5	2.5	4	6	10	15	22	35	54	87	140	220	350	540	870	1400	2200
Over 120																		
To and inc 180	1.2	2	3.5	5	8	12	18	25	40	63	100	160	250	400	630	1000	1600	2500
Over 180																		
To and inc 250	2	3	4.5	7	10	14	20	29	46	72	115	185	290	460	720	1150	1850	2900
Over 250																		
To and inc 315	2.5	4	6	8	12	16	23	32	52	81	130	210	320	520	810	1300	2100	3200
Over 315																		
To and inc 400	3	5	7	9	13	18	25	36	57	89	140	230	360	570	890	1400	2300	3600
Over 400																		
To and inc 500	4	6	8	10	15	20	27	40	63	97	155	250	400	630	970	1550	2500	4000

*Upto 1 mm, Grades 14 to 16 are not provided.

Table 15.2 Fundamental deviations for shafts of types **a** to **k** of sizes upto 500mm (*contd.*)

Fundamental deviation in microns										(1 micron = 0.001 mm)								
Diameter steps in mm		Upper deviation (<i>es</i>)								js*	Lower deviation (<i>ei</i>)							
		<i>a</i>	<i>b</i>	<i>c</i>	<i>d</i>	<i>e</i>	<i>f</i>	<i>g</i>	<i>h</i>		<i>j</i>	<i>k</i>						
over	upto	All grades									5.6	7	8	4 to 7	≤ 3, > 7			
—	*3	− 270	− 140	− 60	− 20	− 14	− 6	− 2	0	± IT/2	− 2	− 4	− 6	− 0	− 0			
3	6	− 270	− 140	− 70	− 30	− 20	− 10	− 4	0		− 2	− 4	—	+ 1	0			
6	10	− 280	− 150	− 80	− 40	− 25	− 13	− 5	0		− 2	− 5	—	+ 1	0			
10	14	− 290	− 150	− 95	− 50	− 32	− 16	− 6	0		− 3	− 6	—	+ 1	0			
14	18																	
18	24										− 4	− 8	—	+ 2	0			
24	30	− 300	− 160	− 110	− 65	− 40	− 20	− 7	0		− 5	− 10	—	+ 2	0			
30	40										− 310	− 170	− 120	− 80	− 50	− 25	− 9	0
40	50										− 320	− 180	− 130					
50	65	− 340	− 190	− 140	− 100	− 60	− 30	− 10	0		− 7	− 12	—	+ 2	0			
65	80	− 360	− 200	− 150														
80	100	− 380	− 220	− 170							− 120	− 72	− 36	− 12	0			
100	120	− 410	− 240	− 180	− 145	− 85	− 43	− 14	0		− 9	− 15	—	+ 3	0			
120	140	− 460	− 260	− 200														
140	160	− 520	− 280	− 210														
160	180	− 580	− 310	− 230														
180	200	− 660	− 340	− 240	− 170	− 100	− 50	− 15	0	± IT/2								
200	225	− 740	− 380	− 260							− 13	− 21	—	+ 4	0			
225	250	− 820	− 420	− 280														
250	280	− 920	− 480	− 300	− 190	− 110	− 56	− 17	0		− 16	− 26	—	+ 4	0			
280	315	− 1050	− 540	− 330	− 210	− 125	− 62	− 18	0		− 18	− 28	—	+ 4	0			
315	355	− 1200	− 600	− 360														
355	400	− 1350	− 680	− 400														
400	450	− 1500	− 760	− 440	− 230	− 135	− 68	− 20	0		− 20	− 32	—	+ 5	0			
450	500	− 1650	− 840	− 480														

*The deviations of shafts of types **a** and **b** are not provided for diameters upto 1 mm

+ For types **js** in the particular Grades 7 to 11, the two symmetrical deviations ± IT/2 may possibly be rounded, if the IT value in microns is an odd value; by replacing it by the even value immediately below.

Table 15.2 for fundamental deviations for shafts (from m to zc)

Fundamental deviation		Lower deviation e_i													
Letter		m	n	p	r	s	t	u	v	x	y	z	za	zb	zc
Grade		01 to 16													
Nominal size															
Over	To														
mm	mm														
-	3	+2	+4	+6	+10	+14	-	+18	-	+20	-	+26	+32	+40	+60
3	6	+4	+8	+12	+15	+19	-	+23	-	+28	-	+35	+42	+50	+80
6	10	+6	+10	+15	+19	+23	-	+28	-	+34	-	+42	+52	+67	+97
10	14	+7	+12	+18	+23	+28	-	+33	-	+40	-	+50	+64	+90	+130
14	18								+39	+45	-	+60	+77	+108	+150
18	24	+8	+15	+22	+28	+35	-	+41	+47	+54	+63	+73	+98	+136	+188
24	30								+48	+55	+64	+75	+88	+118	+160
30	40	+9	+17	+26	+34	+43	+	+48	+60	+68	+80	+94	+112	+148	+200
40	50								+54	+70	+81	+97	+114	+136	+218
50	65	+11	+20	+32	+41	+53	+66	+87	+102	+122	+144	+172	+226	+300	+405
65	80				+43	+59	+75	+102	+120	+146	+174	+210	+274	+360	+480
80	100	+13	+23	+37	+51	+71	+91	+124	+146	+178	+214	+258	+335	+445	+585
100	120				+54	+79	+104	+144	+172	+210	+254	+310	+400	+525	+690
120	140	+15	+27	+43	+63	+92	+122	+170	+202	+248	+300	+365	+470	+620	+800
140	160				+65	+100	+134	+190	+228	+280	+340	+415	+535	+700	+900
160	180				+68	+108	+146	+210	+252	+310	+380	+465	+600	+780	+1000
180	200				+77	+122	+166	+236	+284	+350	+425	+520	+670	+880	+1150
200	225	+17	+31	+50	+80	+130	+180	+258	+310	+385	+470	+575	+740	+960	+1250
225	250				+84	+140	+196	+284	+340	+425	+520	+640	+820	+1050	+1350
250	280	+20	+34	+56	+94	+158	+218	+315	+385	+475	+580	+710	+920	+1200	+1550
280	315				+98	+170	+240	+350	+425	+525	+650	+790	+1000	+1300	+1700
315	355	+21	+37	+62	+108	+190	+268	+390	+475	+590	+730	+900	+1150	+1500	+1900
355	400				+144	+208	+294	+435	+530	+660	+820	+1000	+1300	+1650	+2100
400	450	+23	+40	+68	+126	+232	+330	+490	+595	+740	+920	+1100	+1450	+1850	+2400
450	500				+132	+252	+360	+540	+660	+820	+1000	+1250	+1600	+2100	+2600

Table 15.3 Fundamental deviations for holes of types A to N for sizes up to 500mm

Fundamental deviation in microns										(1 micron = 0.001 mm)														
Diameter steps in mm		Lower deviations (EI)										Upper deviations (ES)												
		A*	*B	C	D	E	F	G	H	J _s +	J			K		M		N						
Over	Upto	All grades										± IT/2	6	7	8	≤ 8	> 8	≤ 8 ‡	> 8	≤ 8	> 8*	≤ 7		
—	3*	+ 270	+ 140	+ 60	+ 20	+ 14	+ 6	+ 2	0	+ 2	+ 4		+ 6	0	0	− 2	− 2	− 4	− 4	Same deviation as for grades > 7 + Δ				
3	6	+ 270	+ 140	+ 70	+ 30	+ 20	+ 10	+ 4	0	+ 5	+ 6		+ 10	− 1 + Δ	—	− 4 + Δ	− 4 + Δ	− 8 + Δ	0					
6	10	+ 280	+ 150	+ 80	+ 40	+ 25	+ 13	+ 5	0	+ 5	+ 8		+ 12	− 1 + Δ	—	− 6 + Δ	− 6 + Δ	− 10 + Δ	0					
10	14	+ 290	+ 150	+ 95	+ 50	+ 32	+ 16	+ 6	0	+ 6	+ 10		+ 15	− 1 + Δ	—	− 7 + Δ	− 7	− 12 + Δ	0					
14	18																							
18	24	+ 300	+ 160	+ 110	+ 65	+ 40	+ 20	+ 7	0	+ 8	+ 12		+ 20	− 2 + Δ	—	− 8 + Δ	− 8	− 15 + Δ	0					
24	30																							
30	40	+ 310	+ 170	+ 120	+ 80	+ 50	+ 25	+ 9	0	+ 10	+ 14		+ 24	− 2 + Δ	—	− 9 + Δ	− 9	− 17 + Δ	0					
40	50	+ 320	+ 180	+ 130																				
50	65	+ 340	+ 190	+ 140	+ 100	+ 60	+ 30	+ 10	0	+ 13	+ 18	+ 28	− 2 + Δ	—	− 11 + Δ	− 11	− 20 + Δ	0						
65	80	+ 360	+ 200	+ 150																				
80	100	+ 380	+ 220	+ 170	+ 120	+ 72	+ 36	+ 12	0	+ 16	+ 22	+ 34	− 3 + Δ	—	− 13 + Δ	− 13	− 23 + Δ	0						
100	120	+ 410	+ 240	+ 180																				
120	140	+ 460	+ 260	+ 200	+ 145	+ 85	+ 43	+ 14	0	± IT/2	+ 18	+ 26	+ 41	− 3 + Δ	—	− 15 + Δ	− 15	− 27 + Δ	0	Same deviation as for grades > 7 + Δ				
140	160	+ 520	+ 280	+ 210																				
160	180	+ 580	+ 310	+ 230																				
180	200	+ 660	+ 340	+ 240	+ 170	+ 100	+ 50	+ 15	0		+ 22	+ 30	+ 47	− 4 + Δ	—	− 17 + Δ	− 17	− 31 + Δ	0					
200	225	+ 740	+ 380	+ 260																				
225	250	+ 820	+ 420	+ 280																				
250	280	+ 920	+ 480	+ 300	+ 190	+ 110	+ 56	+ 17	0		+ 25	+ 36	+ 55	− 4 + Δ	—	− 20 + Δ	− 20	− 34 + Δ	0					
280	315	+ 1050	+ 540	+ 330																				
315	355	+ 1200	+ 600	+ 360	+ 210	+ 125	+ 62	+ 18	0		+ 29	+ 39	+ 60	− 4 + Δ	—	− 21 + Δ	− 21	− 37 + Δ	0					
355	400	+ 1350	+ 680	+ 400																				
400	450	+ 1500	+ 760	+ 440																				
450	500	+ 1650	+ 840	+ 480	+ 230	+ 135	+ 68	+ 20	0	+ 33	+ 43	+ 66	− 5 + Δ	—	− 23 + Δ	− 23	− 40 + Δ	0						

* The deviation of holes of types A and B in all grades >8 are not for diameters upto 1 mm.

+ For the hole of type J_s in the grades 7 and 11, the two symmetrical ± deviations IT/2 may possibly rounded. If the IT value in microns is an odd value, replace it by the even value immediately below.

‡ Special case: For the hole M6, ES = 9 from 250 to 315 (instead of − 11).

Table 15.3: for fundamental deviations for holes (from P to ZC).

Fundamental deviation		Upper deviation ES												Values for Δ^*							
Letter	P to ZC	P	R	S	T	U	V	X	Y	Z	ZA	ZB	ZC								
Grade		>7												Grades:							
Nominal sizes																					
Over	To	-	-	-	-	-	-	-	-	-	-	-	-	3	4	5	6	7	8		
mm	mm																				
-	3	6	10	14	-	18	-	20	-	26	32	40	60	0	0	0	0	0	0		
3	6	12	15	19	-	23	-	28	-	35	42	50	80	1	1.5	1	3	4	6		
6	10	15	19	23	-	28	-	34	-	42	52	67	97	1	1.5	2	3	6	7		
10	14	18	23	28	-	33	-	40	-	50	64	90	130	1	2	3	3	7	9		
14	18						39	45	-	60	77	108	150								
18	24	22	28	35	-	41	47	54	63	73	98	136	188	1.5	2	3	4	8	12		
24	30				41	48	55	64	75	88	118	160	218								
30	40	26	34	43	48	60	68	80	94	112	148	200	274	1.5	3	4	5	9	14		
40	50				54	70	81	97	114	136	180	242	325								
50	65	32	41	53	66	87	102	122	144	172	226	300	405	2	3	5	6	11	16		
65	80		43	59	75	102	120	146	174	210	274	360	480								
80	100	37	51	71	91	124	146	178	214	258	335	445	585	2	4	5	7	13	19		
100	120		54	79	104	144	172	210	254	310	400	525	690								
120	140		63	92	122	170	202	248	300	365	470	620	800								
140	160	43	65	100	134	190	228	280	340	415	535	700	900	3	4	6	7	15	23		
160	180		68	108	146	210	252	310	380	465	600	780	1000								
180	200		77	122	166	266	284	350	425	520	670	880	1150								
200	225	50	80	130	180	258	310	385	470	575	740	960	1250	3	4	6	9	17	26		
225	250		84	140	196	284	340	425	520	640	820	1050	1350								
250	280		94	158	218	315	385	475	580	710	920	1200	1550								
280	315	56	98	170	240	350	425	525	650	790	1000	1300	1700	4	4	7	9	20	29		
315	355																				
355	400	62	108	190	268	390	475	590	730	900	1150	1500	1800	4	5	7	11	21	32		
400	450		114	208	294	435	530	660	820	1000	1300	1650	2100								
450	500	68	126	232	330	490	595	740	920	1100	1450	1850	2400	5	5	7	13	23	34		
			132	252	360	540	660	820	1000	1250	1600	2100	2600								

Table 15.4: Formulae for fundamental deviation for shafts up to 500 mm

<i>Upper deviation (es)</i>		<i>Lower deviation (ei)</i>	
<i>Shaft designation</i>	<i>In microns (for D in mm)</i>	<i>Shaft designation</i>	<i>In microns (For D in mm)</i>
a	$= -(265 + 1.3D)$ for $D \leq 120$	k4 to k7	$= 0.6 \sqrt[3]{D}$
	$= -3.5 D$ for $D > 120$	k for grades ≤ 3 and ≥ 8	$= 0$
b	$\approx -(140 + 0.85 D)$ for $D \leq 160$	m	$= + (IT 7 - IT 6)$
	$\approx -1.8 D$ for $D > 160$	n	$= + 5 D^{0.34}$
c	$= -52 D^{0.2}$ for $D \leq 40$	p	$= + IT 7 + 0 \text{ to } 5$
	$= -(95 + 0.8 D)$ for $D > 40$	r	$= \text{geometric mean of values } ei \text{ for } p \text{ and } s$
d	$= -16 D^{0.44}$	s	$= + IT 8 + 1 \text{ to } 4$ for $D \leq 50$
e	$= -11 D^{0.41}$		$= + IT 7 + 0.4 D$ for $D > 50$
f	$= -5.5 D^{0.41}$	t	$= IT 7 + 0.63 D$
g	$= -2.5 D^{0.34}$	u	$= + IT 7 + D$
h	$= 0$	v	$= + IT 7 + 1.25 D$
		x	$= + IT 7 + 1.6 D$
		y	$= + IT 7 + 2 D$
		z	$= + IT 7 + 2.5 D$
		za	$= + IT 8 + 3.15 D$
		zb	$= + IT 9 + 4 D$
j5 to j8	no formula	zc	$= + IT 10 + 5 D$